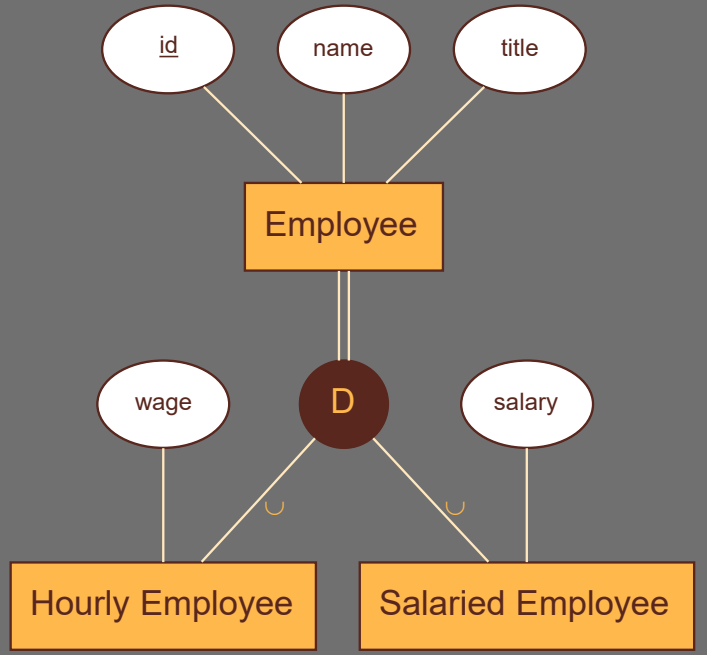
ERD Implementation: Classes & Categories

Fundamental Problems

Able to express complex constraints:

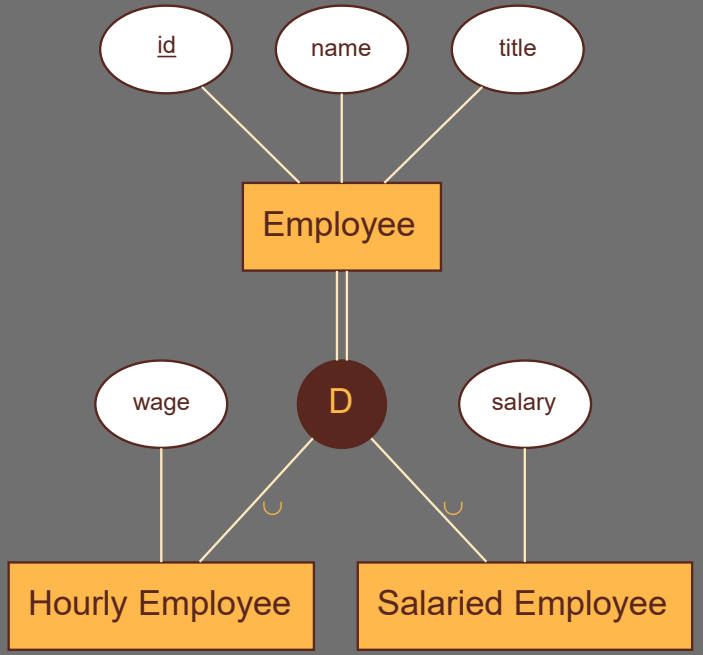
- Superclass = subclass union
- Subclasses disjoint



Fundamental Problems

Inheritance:

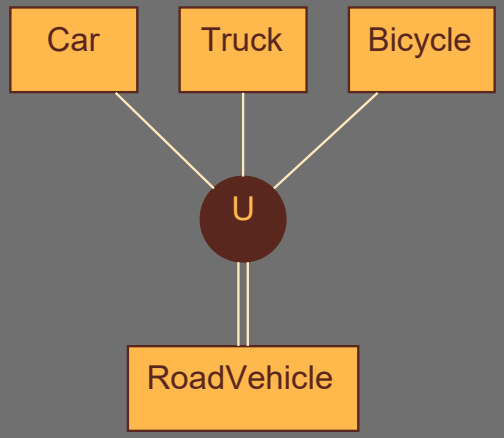
- Entity attributes can be in multiple entity sets.



Fundamental Problems

Inheritance:

- Entity attributes can be in multiple entity sets.
- Attributes within entity set vary due to categories.





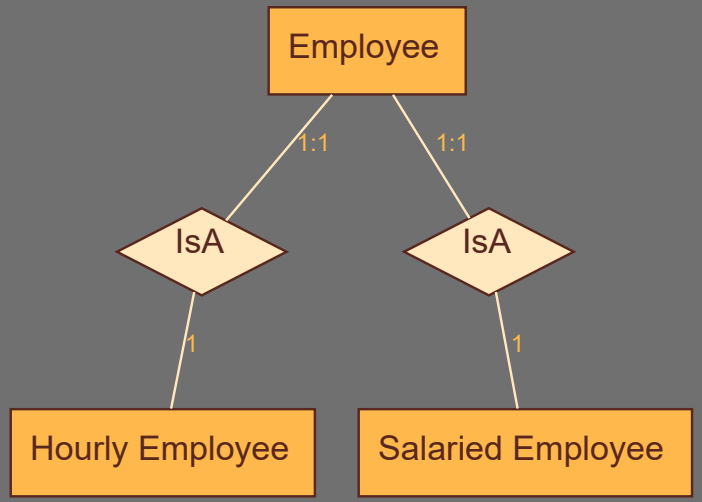
Dealing with super/subclasses is one of the relational model's main weaknesses.

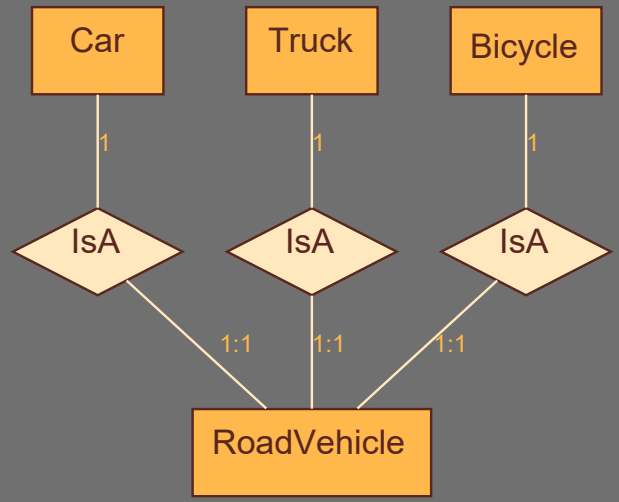
Trade-off



Semantic accuracy

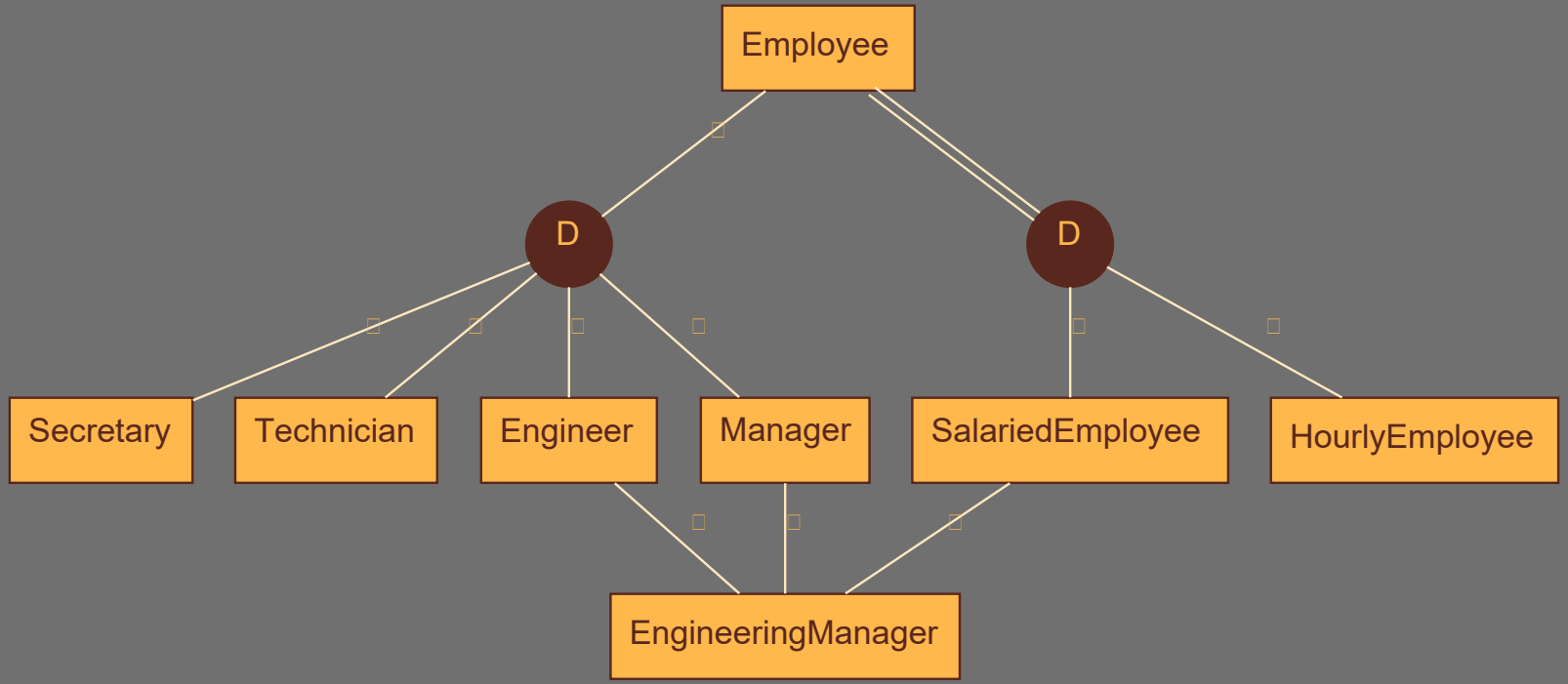
Implement O D U as 1-1 relationship X-IsA-Y.

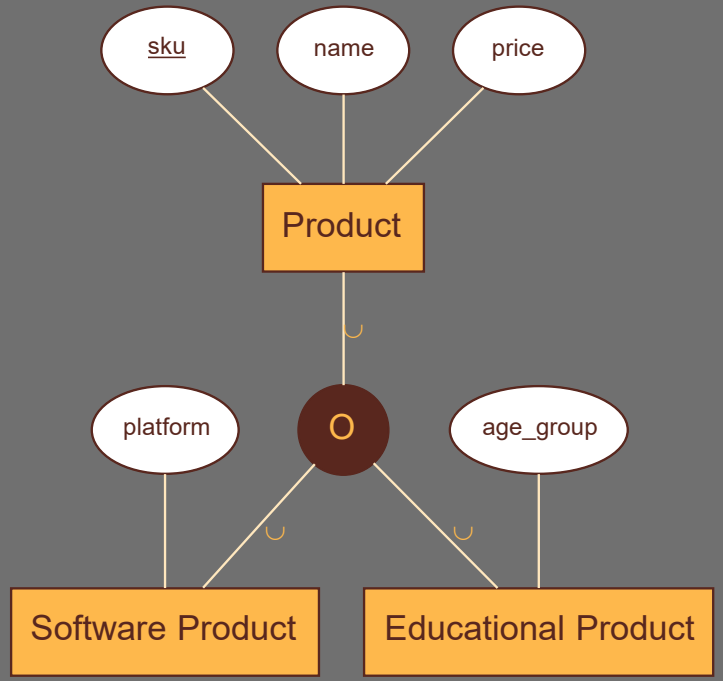




Simplicity & efficiency

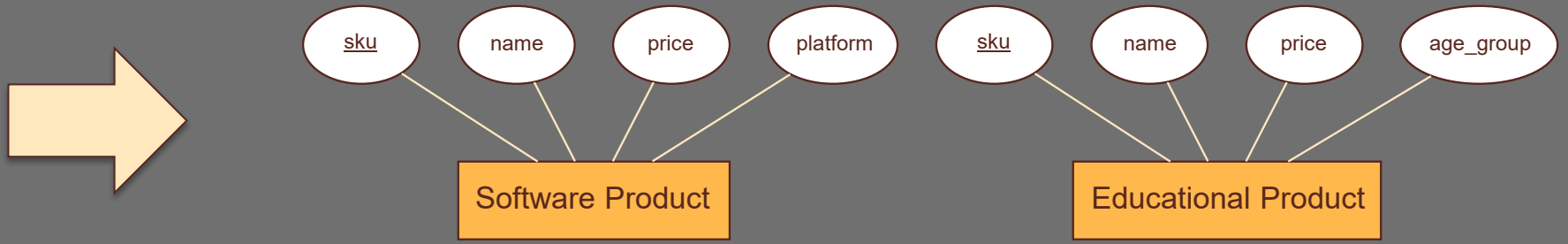
Collapse hierarchy to eliminate inheritance.

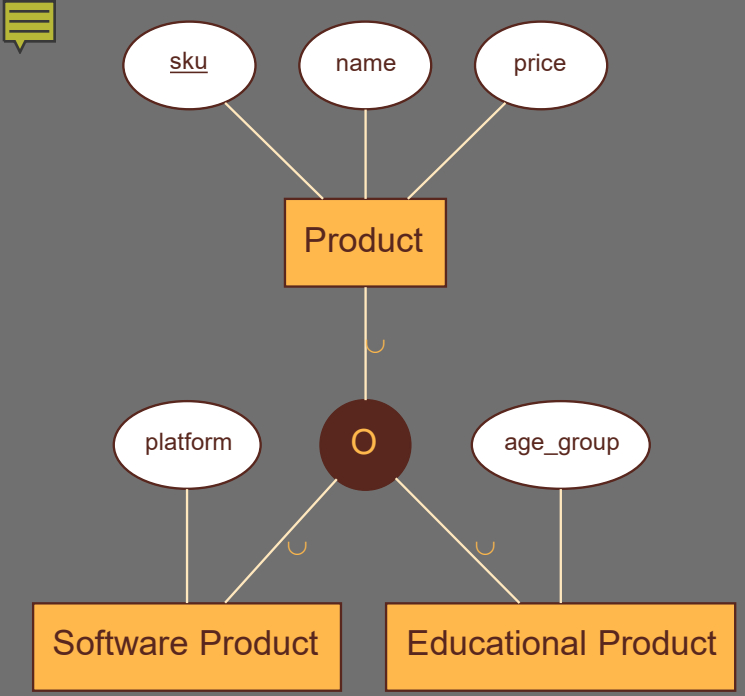




Can duplicate data

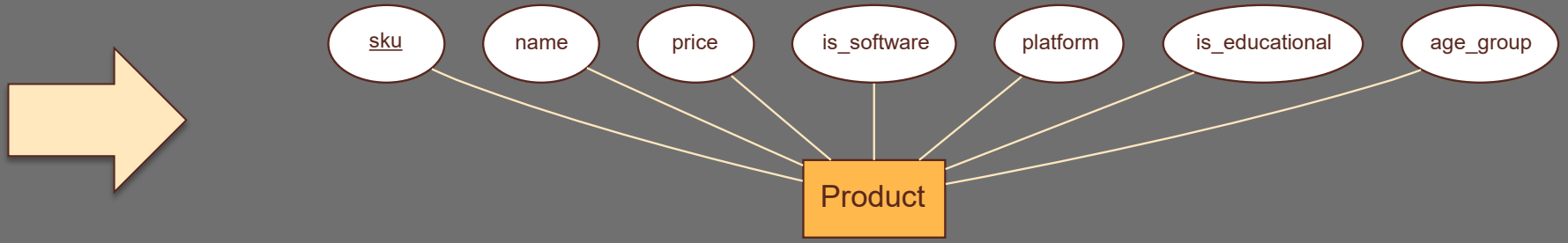
Duplicates data in both subclasses.
Duplicates relations that were on superclass.

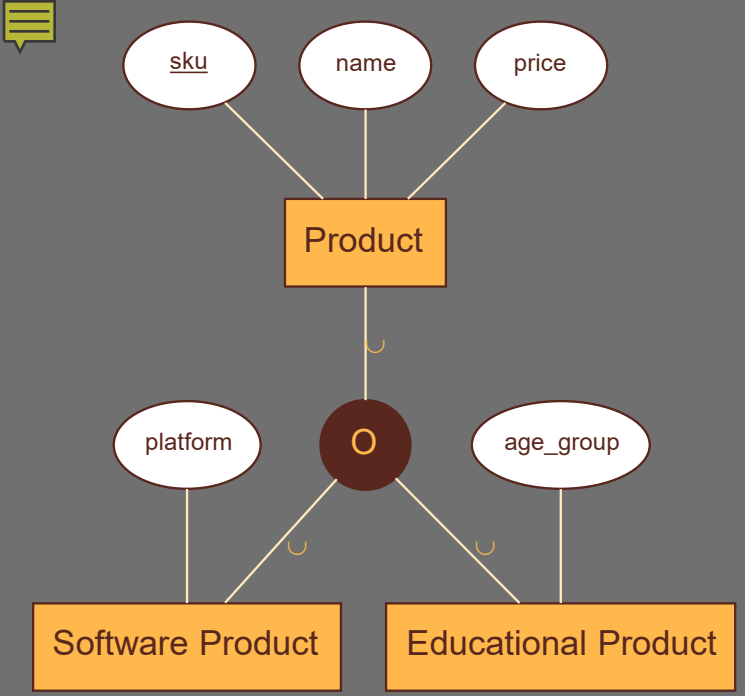




Can add attributes

Need to know subclass for each entity.





Can be many NULLs

Not all attributes relevant for all entities.

